mergencystretch

Chapter 9. Regression Discontinuity, Difference-in-Differences and Synthetic Control

Sharp and fuzzy RD, DiD and staggered adoption, synthetic control and synthetic DiD

Contents

9.1 Regression discontinuity **2**
- 9.1.1 Where RDD fits . . . 2
- 9.1.2 Concept and identification . . . 2
  - 9.1.2.1 Examples . . . 2
  - 9.1.2.2 The design . . . 3
  - 9.1.2.3 Estimand and identification . . . 3
- 9.1.3 Validity and diagnostics . . . 4
- 9.1.4 Estimation . . . 4
  - 9.1.4.1 General strategy . . . 4
  - 9.1.4.2 Local linear regression . . . 5
  - 9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth . . . 5
  - 9.1.4.4 Why the MSE-optimal bandwidth may fail for inference . . . 6
  - 9.1.4.5 Covariates and software . . . 6
- 9.1.5 Fuzzy RD . . . 6
- 9.1.6 Case study: remedial education in Chicago . . . 7
- 9.1.7 Summary . . . 7
- 9.1.8 Appendix: Kink RD . . . 8
- 9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth . . . 8
- 9.1.10 Appendix: Regularization of the bandwidth . . . 8
- 9.1.11 Appendix: Why local linear? Polynomial order . . . 8
- 9.1.12 Appendix: Global polynomials: a cautionary tale . . . 9

9.2 Difference-in-differences **9**
- 9.2.1 Motivation . . . 9
  - 9.2.1.1 Case study: minimum wages (Card and Krueger 1994) . . . 9
- 9.2.2 The DiD design . . . 9
  - 9.2.2.1 Setup . . . 9
  - 9.2.2.2 The identification problem . . . 10
  - 9.2.2.3 Three control strategies . . . 10
  - 9.2.2.4 Parallel trends . . . 10
  - 9.2.2.5 Estimation . . . 11
  - 9.2.2.6 DiD and two-way fixed effects (2×2) . . . 11
- 9.2.3 Conditional and nonlinear DiD . . . 11
  - 9.2.3.1 Conditional parallel trends . . . 12
  - 9.2.3.2 Nonlinear DiD (changes-in-changes) . . . 12
  - 9.2.3.3 Key takeaways . . . 12
- 9.2.4 Appendix: Pre-testing bias . . . 13
- 9.2.5 Appendix: Sensitivity analysis for parallel trends . . . 13
- 9.2.6 Appendix: Staggered treatment timing . . . 13

9.3 Staggered difference-in-differences **13**
- 9.3.1 Staggered adoption . . . 13
  - 9.3.1.1 TWFE as is is not sufficient . . . 14

9.3.1.2 Terminology . . . 14
9.3.1.3 Valid and invalid comparisons . . . 14
9.3.2 Callaway–Sant'Anna group-time ATTs . . . 14
9.3.2.1 Notation . . . 14
9.3.2.2 Parallel trends . . . 15
9.3.2.3 Group-time ATT and identification . . . 15
9.3.2.4 Conditional parallel trends . . . 15
9.3.2.5 Aggregation . . . 16
9.3.2.6 Summary . . . 16

9.4 Synthetic control **16**
9.4.1 From DiD to synthetic controls . . . 16
9.4.1.1 Case study: California Proposition 99 . . . 17
9.4.2 The estimator . . . 17
9.4.2.1 Setup . . . 17
9.4.2.2 Missing counterfactuals and synthetic weights . . . 17
9.4.2.3 Model-based justifications . . . 17
9.4.2.4 Convexity: no extrapolation . . . 18
9.4.2.5 Practical concerns . . . 18
9.4.3 Inference and bias . . . 18
9.4.3.1 Placebo tests . . . 19
9.4.3.2 Why imperfect pre-treatment fit causes bias . . . 19
9.4.3.3 Bias correction: augmented SCM . . . 19
9.4.4 Discussion . . . 19
9.4.5 Appendix: More treated units . . . 20
9.4.6 Appendix: Predictor importance weights V . . . 21

9.5 Synthetic difference-in-differences **21**
9.5.1 DiD, SC and SDiD . . . 21
9.5.2 Formulations . . . 21
9.5.2.1 Setting and algorithm . . . 22
9.5.2.2 Unit weights . . . 22
9.5.2.3 Time weights . . . 22
9.5.3 Comparisons and takeaways . . . 23

9.1 Regression discontinuity

Motivation. The central question is how a known cutoff rule for treatment assignment identifies a causal effect at the cutoff, how that effect is estimated from data near the boundary, and what the bandwidth choice does to bias, variance and inference.

9.1.1 Where RDD fits

Motivation. Place RDD among empirical designs.

Types of empirical studies, from more to less researcher control:

Experiment (RCT)	Natural experiment	Quasi-experiment	Observational study
researcher randomizes treatment	as-if random assignment by nature/policy (lottery, IV)	known non-random rule exploited (RD, DiD, SCM)	confounders must be assumed away (matching, IPW)

Method	Design type	Key source of variation
Back-door / matching / IPW	observational	adjusted confounders
IV	natural / quasi-experiment	exogenous instrument
RDD	quasi-experiment	cutoff rule
DiD	quasi-experiment	temporal variation across groups
SCM	quasi-experiment	weighted donor pool

IV straddles the boundary: the instrument may come from a natural experiment (lottery) or a quasi-experimental rule (policy threshold). From here on the focus is on quasi-experimental designs that exploit known institutional rules. So far: back-door adjustment (block non-causal paths) and IV (external instrument under unobserved confounding). Today: treatment determined by whether a pre-treatment variable crosses a known threshold.

Notation. $Z_i \in \mathbb{R}$: forcing (running) variable (not an instrument as in Chapter 7); c: cutoff; $X_i = \mathbb{1}\{Z_i \geq c\}$: treatment in the sharp design; $Y_i(1), Y_i(0)$: potential outcomes; τ_{SRD}: sharp-RD effect at the cutoff; $W_i = \mathbb{1}\{Z_i \geq c\}$: the instrument in the fuzzy design (playing the role of Z_i in the IV lecture).

9.1.2 Concept and identification

Motivation. Define the sharp RD effect and show that continuity identifies it.

9.1.2.1 Examples

Incumbency advantage (Lee 2008; Lee et al. 2004). Does winning an election raise the chance of winning the next? Running variable: Democratic vote-share margin at election t; cutoff: 50%; outcome: Democratic vote share at $t+1$ (or the ADA score of the representative). Candidates who barely win and barely lose are comparable in observables and unobservables; near the cutoff the election outcome is as good as random. Barely winning causes a 6–8 percentage-point increase in next-election vote share: a substantial incumbency advantage. Often regarded as the gold-standard RDD application.

Scholarships and test scores (Thistlethwaite and Campbell 1960, the first RDD): eligibility depends solely on whether an exam score exceeds a threshold; students just above and below are

arguably comparable in ability and attitudes. **Financial aid and family income**: eligibility depends on family income below a threshold. In both, the cutoff is known and pre-determined and units near it are plausibly comparable.

9.1.2.2 The design

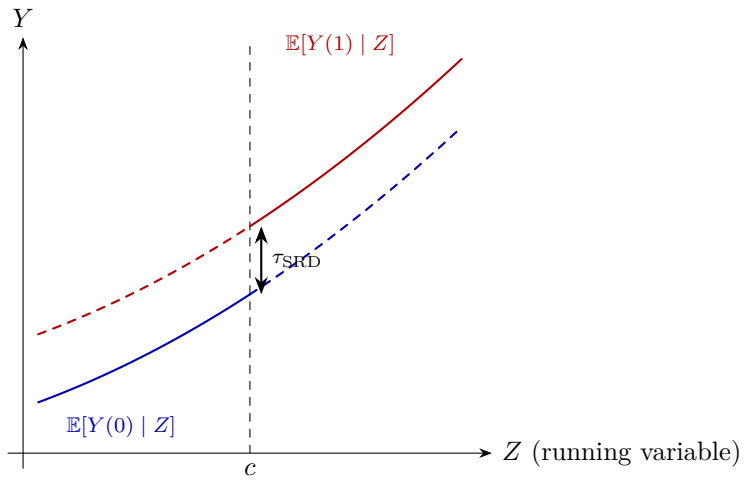


Figure. *Sharp RD: units with $Z \geq c$ are treated. We observe $\mathbb{E}[Y(0) \mid Z]$ only to the left of c and $\mathbb{E}[Y(1) \mid Z]$ only to the right (solid). If both are continuous at c, the jump of the observed regression function at c is the effect at the cutoff.*

In an RDD, treatment status changes discontinuously in a pre-treatment forcing variable, while unobserved confounders vary smoothly around the cutoff. Knowledge of the assignment mechanism identifies a *local* causal effect at the cutoff, by comparing units with similar forcing values but different treatment. The discontinuity is often created by an artificial, pre-fixed threshold.

Sharp RD. Z_i continuous (discrete running variables need extra care; Cattaneo et al. 2020) and $X_i = \mathbb{1}\{Z_i \geq c\}$, a deterministic function of the forcing variable (scores above 1500 get the scholarship). Positivity is violated by construction:

$$P[X_i = 1 \mid Z_i = c - \varepsilon] = 0, \qquad P[X_i = 1 \mid Z_i = c + \varepsilon] = 1,$$

so back-door adjustment and IPW cannot deliver the ATE/ATT; continuity-based extrapolation at the cutoff is used instead.

Fuzzy RD. Crossing the threshold changes the *probability* of treatment discontinuously, but not from 0 to 1 (Section 9.1.5).

9.1.2.3 Estimand and identification

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] = \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c].$$

Since Z_i is continuous, no unit has $Z_i = c$ exactly; identification relies on local extrapolation, which needs an assumption.

Continuity assumption. The CEFs $\mu_1(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_0(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$ are continuous in z. Then

$$\begin{aligned}
\mathbb{E}[Y_i(0) \mid Z_i = c] &= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid Z_i = z] && \text{(continuity)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid X_i = 0, Z_i = z] && \text{(SRD: } z < c \Rightarrow X_i = 0\text{)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] && \text{(consistency),}
\end{aligned}$$

and likewise $\mathbb{E}[Y_i(1) \mid Z_i = c] = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Hence

$$\boxed{\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z].}$$

The estimation problem is to estimate two regression functions at a boundary point: very hard without parametric assumptions; nonparametric regression is consistent but converges slowly.

What can go wrong? Sorting around the threshold violates smoothness: students retake an exam to pass the aid threshold; richer students retake more often, so those just above differ systematically from those just below.

9.1.3 Validity and diagnostics

Motivation. Check the design's assumptions graphically and with falsification tests.

Most RD analyses show some of three figure types:

1. **Outcomes by running variable:** is there a jump at the cutoff? "An RDD is hard to sell if the effect is not visually obvious in the outcome plot" (Imbens and Lemieux 2008).
2. **Covariates by running variable:** covariates should be smooth across c; a discontinuity in covariate means suggests discontinuities in potential-outcome means, and the RD may be invalid.
3. **Density of the running variable:** is there evidence of sorting? Estimate the density separately on each side; a discontinuity suggests sorting.

Manipulation and the McCrary test. RDD assumes units cannot precisely manipulate Z_i to land on a preferred side. If they can, smoothness of potential outcomes is likely violated. McCrary (2008) tests for a discontinuity of the density $f_Z(z)$ at c; a significant jump suggests manipulation, in which case units just above and below are no longer comparable.

Bunching. With administrative data, individuals or firms are often seen to locate at policy thresholds: firms reporting earnings just below a tax or regulation threshold, workers choosing hours just below the full-time cutoff (World Bank blog "Ready, set, bunch"). Example: the distribution of reported wealth in Colombia before (2009) and after (2010) a wealth-tax reform shows bunching just below the new threshold (Londoño-Vélez).

Falsification (placebo) tests. If the RDD is valid, the effect appears only at the true cutoff.

- *Placebo cutoffs:* re-estimate at fake cutoffs away from c; with smooth CEFs no jump should appear. In a simulation with an effect only at $c = 0$, estimates at placebo cutoffs between -2 and 2 are near zero with CIs covering zero, while the true cutoff shows a clear effect.
- *Placebo outcomes:* use pre-treatment covariates as outcomes; treatment cannot affect them, so a jump signals sorting.

9.1.4 Estimation

Motivation. Estimate boundary limits with local polynomial regression and choose the bandwidth.

9.1.4.1 General strategy

The goal is limits such as $\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Two differences from standard nonparametric regression: the target is a single point, and it is a *boundary* point where data exist on one side only, so boundary bias shrinks slowly. The naive difference in means uses data too far from the boundary.

Upper and lower limits: $\mu_+(c) = \lim_{z\downarrow c}\mathbb{E}[Y_i \mid Z_i = z]$, $\mu_-(c) = \lim_{z\uparrow c}\mathbb{E}[Y_i \mid Z_i = z]$, and $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

Kernel (local average) regression. With a uniform kernel and bandwidth h,

$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{c-h \le Z_i < c\}}{\sum_{i=1}^n \mathbb{1}\{c-h \le Z_i < c\}},$$

the mean among units at most h below the threshold (similarly above). Equivalently, regress on units within h of c:

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg\min_{\alpha,\tau} \sum_{i:Z_i\in[c-h,c+h]} (Y_i - \alpha - \tau X_i)^2,$$

where, with binary X_i, the OLS coefficient is the difference in means. This assumes locally constant potential outcomes on each side. h governs the bias–variance trade-off: large h, high bias and low variance; small h, low bias and high variance. Bias shrinks slowly in h, so finite-sample bias may be large. Kernels: uniform $K(u) = \frac{1}{2}\mathbb{1}\{|u| < 1\}$, triangular $K(u) = (1 - |u|)\mathbb{1}\{|u| < 1\}$.

9.1.4.2 Local linear regression

With a wider bandwidth the CEF may not be flat, so allow a slope. On $Z_i \in [c-h, c)$,

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg\min_{\alpha,\beta} \sum_{i:Z_i\in[c-h,c)} \big(Y_i - \alpha - \beta(Z_i - c)\big)^2,$$

and the same on $[c, c+h]$. Then

$$\hat{\tau}_{\text{SRD}} = \hat{\mu}_+(c) - \hat{\mu}_-(c) = \hat{\alpha}_+ + \hat{\beta}_+(c-c) - \hat{\alpha}_- - \hat{\beta}_-(c-c) = \hat{\alpha}_+ - \hat{\alpha}_-.$$

Single regression. Equivalently,

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i:Z_i\in[c-h,c+h]} \{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\}^2,$$

with $\hat{\tau}_{\text{SRD}} = \hat{\tau}$. For $X_i = 0$ the fit is $\alpha + \beta(Z_i - c)$; for $X_i = 1$ it is $(\alpha + \tau) + (\beta + \gamma)(Z_i - c)$: one regression on top of the other.

Kernel weighting. Weight points near c more heavily:

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i=1}^n K\left(\frac{Z_i - c}{h}\right) \{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\}^2,$$

typically with the triangular kernel.

9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth

Let $\mu(z) = \mathbb{E}[Y \mid Z = z]$ with one-sided second derivatives $\mu_\pm^{(2)}(c)$, density $f(c)$ of Z at c, and $\sigma_\pm^2(c) = \text{Var}(Y \mid Z = c^\pm)$. For local linear regression with bandwidth h,

$$\text{Bias} \approx C_1 h^2 \underbrace{\big(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\big)}_{\text{curvature difference}}, \qquad \text{Variance} \approx \frac{C_2}{Nh} \cdot \underbrace{\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c)}}_{\text{noise / density at } c}.$$

As h grows the variance falls like $1/h$ but the bias grows like h^2; as h shrinks the bias falls but the variance explodes. Minimizing MSE = Bias^2 + Variance (Imbens and Kalyanaraman 2012):

$$h_{\text{MSE}} = C_K \left(\frac{\sigma_+^2(c)/f_+(c) + \sigma_-^2(c)/f_-(c)}{\big(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\big)^2}\right)^{1/5} N^{-1/5}.$$

(Setting the derivative of $C_1^2 h^4 D^2 + C_2 V/(Nh)$ to zero gives $h^5 \propto V/(ND^2)$.) The optimal bandwidth shrinks with N, but slowly. All quantities (curvatures, variances, density) are unknown and estimated, curvature by local quadratic regression.

9.1.4.4 Why the MSE-optimal bandwidth may fail for inference

h_{MSE} minimizes point-estimation error only. Keeping variance low, it picks a relatively wide window ($\propto N^{-1/5}$), so the bias is not negligible relative to the standard error: a nominal 95% CI may cover only about 80% of the time (undercoverage). Remedies: *robust bias correction* (RBC; Calonico, Cattaneo and Titiunik 2014) re-centers the interval by subtracting an estimate of the leading bias term (and accounts for its variability); the *CE-optimal bandwidth* (Calonico, Cattaneo and Farrell 2020) minimizes coverage error, the gap between nominal and actual coverage, with $h_{\text{RBC}} \propto N^{-1/4}$ for local linear fits. Rule of thumb $\hat{h}_{\text{RBC}} \approx (0.7\text{–}0.8) \times \hat{h}_{\text{MSE}}$: valid p-values need a tighter window than the point estimate.

Bandwidth	Rate	Bias	Variance	CI length	Coverage
MSE-optimal	$N^{-1/5}$	high	low	short	fails
Undersmoothing	$N^{-1/3}$	very low	high	too long	valid
CE-optimal	$N^{-1/4}$	moderate	moderate	efficient	optimal

CE-optimal with RBC is the default in modern software (`rdrobust`).

Bandwidth sensitivity. A credible RDD shows stable estimates across a range of bandwidths. In a simulation with true $\tau = 3$, small h gives low bias and high variance (wide CIs, few observations in the window), large h gives high bias and low variance.

9.1.4.5 Covariates and software

Covariates are not needed for identification (continuity at the cutoff is the identifying restriction, unlike back-door adjustment), but they can improve precision:

$$Y_i = \alpha + \beta(Z_i - c) + \tau X_i + \gamma(Z_i - c)X_i + \delta^\top V_i + \varepsilon_i,$$

with V_i pre-treatment covariates (Calonico et al. 2019). Covariates also serve as diagnostics: a jump in V_i at c is evidence against validity.

`rdrobust` (Calonico, Cattaneo, Titiunik; R/Stata) implements MSE- and CE-optimal bandwidths (default CE-optimal with RBC), robust bias-corrected CIs, covariate adjustment, clustering and fuzzy RD. In R: `library(rdrobust); rd <- rdrobust(Y, Z, c = 1500); summary(rd)`.

9.1.5 Fuzzy RD

Motivation. Treat imperfect compliance with the cutoff as a local instrumental-variable problem.

Concept. In the fuzzy design crossing the cutoff only changes the probability of treatment:

$$\lim_{z \downarrow c} P[X_i = 1 \mid Z_i = z] \neq \lim_{z \uparrow c} P[X_i = 1 \mid Z_i = z],$$

but the jump is not from 0 to 1. Reasons: exemptions and discretionary enforcement, imperfect compliance, administrative errors. Examples: a test-score cutoff determines summer-school eligibility, but not all eligible students attend; age-based eligibility with early or late applicants. Many real RD settings are fuzzy.

Fuzzy RD as IV: the Wald estimator. Take $W_i = \mathbb{1}\{Z_i \geq c\}$ as instrument for the treatment received X_i; near c, W_i is as good as randomly assigned. Under continuity,

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]} = \frac{\text{ITT effect on } Y}{\text{first-stage effect on } X},$$

the Wald estimator, i.e. IV/LATE applied locally at the cutoff.

Assumptions. (1) Continuity of $\mathbb{E}[Y_i(x) \mid Z_i = z]$ and $\mathbb{E}[X_i(w) \mid Z_i = z]$ at c;

(2) first stage: $\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z] \neq 0$;

(3) monotonicity: crossing c makes no one less likely to be treated (no defiers). These mirror relevance, exclusion and monotonicity; exclusion is built in, since W_i affects Y_i only through X_i for compliers near c.

Interpretation. $\tau_{\text{FRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}, Z_i = c]$, a LATE with *double locality*: it applies only to compliers (units whose treatment is actually changed by crossing the cutoff) and only at the cutoff. External validity is limited, internal validity strong; many policy questions naturally produce fuzzy RD settings.

9.1.6 Case study: remedial education in Chicago

Motivation. See a fuzzy RD in action (Jacob and Lefgren 2004, REStat).

Background. In 1996 Chicago Public Schools ended "social promotion": students had to meet minimum scores on the Iowa Test of Basic Skills (ITBS) reading and math tests to advance (e.g. third graders needed 2.8 grade equivalents in both). Students below the cutoff had to attend summer school and could be retained. *Question:* what is the causal effect of summer school and grade retention on achievement?

Design. Treatment: summer school and retention. Fuzzy, since exemptions and discretionary enforcement mean some students above the cutoff are retained and some below are not; but the probability of treatment jumps sharply at the cutoff, identifying the LATE there. Data: CPS administrative records 1993–1999, about 350,000 third and sixth graders; outcomes: changes in ITBS reading and math (in logits) one and two years later.

IV estimation. With the instrument $W_i = \mathbb{1}\{\text{Score}_i \geq \text{cutoff}\}$ (above/below the cutoff; here falling below triggers treatment, so the first-stage jump is negative),

$$\begin{aligned} \text{First stage:} \quad & \text{Treat}_i = \alpha_0 + \alpha_1 W_i + \alpha_2 \text{Score}_i + \alpha_3 (W_i \times \text{Score}_i) + \nu_i, \\ \text{Second stage:} \quad & Y_i = \beta_0 + \tau \widehat{\text{Treat}}_i + \beta_1 \text{Score}_i + \beta_2 (W_i \times \text{Score}_i) + \varepsilon_i, \end{aligned}$$

where τ is the LATE near the cutoff. Relevance: the cutoff strongly predicts treatment; exogeneity: near the threshold, being just above or below is effectively random. (Modern RD software uses local linear regression with $W_i \times \text{Score}_i$ interactions to allow different slopes on each side.)

Results. Net effect (summer school + retention): for third graders roughly 0.11–0.13 logits in the first year (about 20% of a year's typical learning), persisting after two years with some fade-out; for sixth graders no meaningful reading effect and a small, insignificant positive math effect. Retention alone: short-term reading and math gains in grade 3; negative reading and negligible math effects in grade 6. Summer school alone: positive for third graders, especially low performers. Visually, the probability of summer school/retention drops sharply at the promotion cutoff while next-year achievement also drops there, indicating a net positive effect for third graders.

Strengths and limitations. Strengths: a natural experiment via a policy cutoff; fuzzy RD with IV enables causal inference; heterogeneity by grade, gender and SES. Limitations: LATE only (students near the cutoff, compliers); no evidence on long-term (3+ year) outcomes; manual bandwidth choice without MSE- or CE-optimal selection. The study illustrates both the power of fuzzy RD and the importance of modern bandwidth methods.

9.1.7 Summary

- RDD exploits a known cutoff rule for treatment assignment.

- Identification: continuity of the potential-outcome CEFs at the cutoff, $\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$.
- Diagnostics: outcome plot, covariate balance, McCrary density test, placebo tests.
- Estimation: local linear regression with kernel weights.
- Bandwidth: MSE-optimal ($N^{-1/5}$) for point estimation, CE-optimal ($N^{-1/4}$) for inference; rule of thumb, shrink h_{MSE} by 20–30%.
- Fuzzy RD = IV at the cutoff, identifying the LATE for compliers at c.
- Case study: positive effects of remedial education for Chicago third graders, little effect for sixth graders.

9.1.8 Appendix: Kink RD

Sharp kink RD looks for discontinuities in the first derivative (slope) of $\mathbb{E}[Y_i \mid Z_i = z]$ at the threshold rather than in its level (Card, Lee, Pei and Weber 2015). Estimation is similar, but local quadratic regression is preferable.

9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth

(Imbens and Kalyanaraman 2012.) (1) $\{Y_i, Z_i\}$ i.i.d.;

(2) the density $f(z)$ is continuous with $f(c) > 0$;

(3) $\mu(z)$ is three times differentiable near c;

(4) K is symmetric, bounded, with compact support;

(5) $\sigma^2(z)$ is continuous near c with $\sigma^2(c) > 0$.

9.1.10 Appendix: Regularization of the bandwidth

The optimal bandwidth has the squared curvature difference in its denominator; if the estimated curvature is near zero, h explodes. IK (2012) add a penalty $\hat{r}$ (the variance of the curvature estimate) to the denominator:

$$\hat{h}_{\text{reg}} \propto \left(\frac{\text{sum of variances}}{f(c)\,[(\text{curvature})^2 + \text{regularization}]} \right)^{1/5},$$

ensuring a finite bandwidth even for flat data. Implementation: estimate $\hat{\mu}_{\pm}^{(2)}(c)$ by local quadratic regressions and set

$$\hat{r}_+ = \frac{2160\,\hat{\sigma}_+^2(c)}{N_{h,+}h^4}, \qquad \hat{r}_- = \frac{2160\,\hat{\sigma}_-^2(c)}{N_{h,-}h^4},$$

so the penalty grows with less data (N_h) or more noise (σ^2).

9.1.11 Appendix: Why local linear? Polynomial order

RD uses local polynomial regression of order p. Higher p reduces bias but increases variance (more parameters from limited data near c); higher p has faster theoretical rates, but the variance cost dominates in finite samples. Division of labour: local linear ($p = 1$) for the main estimate (good boundary bias–variance balance); local quadratic ($p = 2$) for bias estimation in the MSE-optimal bandwidth (IK 2012) and in robust bias correction (CCT 2014). "Estimate with p, correct bias with $p + 1$" is the `rdrobust` default.

9.1.12 Appendix: Global polynomials: a cautionary tale

An older practice fitted global high-order polynomials (cubic, quartic, ...) over the whole range of the running variable. Gelman and Imbens (2019) show this is problematic: estimates are sensitive to observations far from the cutoff, high-order polynomials oscillate wildly (Runge-type behavior), and confidence intervals have poor coverage. Local polynomials fitted within h of c do not suffer from these problems. Modern best practice is local linear estimation with a data-driven bandwidth. (Further reading: *The Effect*; *The Mixtape*.)

9.2 Difference-in-differences

Motivation. The central question is how repeated measurements on treated and untreated groups identify the effect of treatment on the treated when treatment is not randomized, by assuming that the untreated trend is a valid proxy for the treated group's counterfactual trend.

9.2.1 Motivation

Motivation. Introduce the two sources of variation in panel data.

So far good controls came from randomization, similar covariates (matching/weighting), an exogenous instrument (IV), or proximity to a discontinuity (RD). With repeated measurements on the same units there are two sources of variation: within-unit variation over time (before/after) and cross-sectional variation in treatment. A before–after comparison uses the treated group's own past as its counterfactual; difference-in-differences uses the control group's change as the counterfactual change of the treated group (parallel paths).

9.2.1.1 Case study: minimum wages (Card and Krueger 1994)

Does raising the minimum wage affect employment? Classical theory suggests negative effects, but minimum wages are hard to randomize. In 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05 while neighboring Pennsylvania stayed at \$4.25. Employment is observed in both states before and after the increase, focusing on fast-food restaurants in eastern PA and NJ, which have similar prices, wages and products and are most likely to be affected. In the picture, average employment in NJ (treated) and PA (control) is plotted before and after; the dashed counterfactual line for NJ is parallel to PA's, and the ATT is the gap between NJ's actual and counterfactual post-period levels.

9.2.2 The DiD design

Motivation. Formalize the two-group, two-period design and identify the ATT.

9.2.2.1 Setup

Two groups, two periods. Pre-period ($t = 0$): neither group is treated. Post-period ($t = 1$): one group is treated. $G_i = 1$ for units treated at $t = 1$, $G_i = 0$ for never-treated units. Treatment status: $X_{i0} = 0$ for all; $X_{i1} = G_i$.

	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

Potential outcomes. $Y_{it}(x)$ is the potential outcome under treatment x at time t; the individual effect is $Y_{it}(1) - Y_{it}(0)$. Consistency: $Y_{it} = X_{it}Y_{it}(1) + (1 - X_{it})Y_{it}(0)$. Hence $Y_{i0}(0) = Y_{i0}$ for all units; $Y_{i1} = Y_{i1}(1)$ if $G_i = 1$; $Y_{i1} = Y_{i1}(0)$ if $G_i = 0$.

9.2.2.2 The identification problem

$$\text{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{(b)}.$$

(a) is estimable;

(b) is counterfactual: what would the treated group's average have been had it remained untreated?

9.2.2.3 Three control strategies

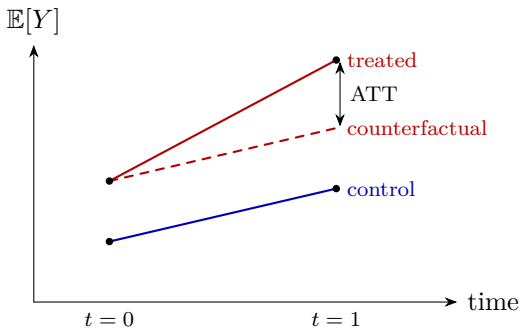


Figure. *Difference-in-differences. Parallel trends assumes that without treatment the treated group would have changed by the same amount as the control group (dashed). The ATT is the treated group's change minus the control group's change.*

1. **Before–after:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1]$ (no change over time absent treatment).
2. **Cross-sectional comparison:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1} \mid G_i = 0]$ (no group differences).
3. **Difference-in-differences:** assume parallel trends; then

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i0} \mid G = 1]) - (\mathbb{E}[Y_{i1} \mid G = 0] - \mathbb{E}[Y_{i0} \mid G = 0]).$$

9.2.2.4 Parallel trends

Definition (Parallel trends)

$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$.

The control group's trend proxies how the treated group would have changed without treatment. Why is this weaker than other assumptions? It accounts for time-invariant unobserved differences between groups and allows for common trends over time. But it is *not invariant to nonlinear transformations*: parallel trends in Y_{it} imply non-parallel trends in $\log Y_{it}$, and vice versa.

Theorem 9.1 (Identification)

Under consistency and parallel trends,

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0]).$$

Proof [Supplement]. By parallel trends, $\mathbb{E}[Y_{i1}(0) \mid G=1] = \mathbb{E}[Y_{i0}(0) \mid G=1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G=0]$. Consistency gives $Y_{i0}(0) = Y_{i0}$ for everyone and $Y_{i1}(0) = Y_{i1}$ for controls, so $\mathbb{E}[Y_{i1}(0) \mid G=1] = \mathbb{E}[Y_{i0} \mid G=1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G=0]$. Substitute into $\text{ATT} = \mathbb{E}[Y_{i1} \mid G=1] - \mathbb{E}[Y_{i1}(0) \mid G=1]$. ■

Threats: non-parallel trends, i.e. unmeasured time-varying confounding. *Ashenfelter's dip:* earnings often fall temporarily just before entry into a training program. The dip makes the treated group's pre-period outcome artificially low, inflating its before–after difference, so the effect is overestimated. *Falsification test:* check parallel trends in pre-treatment periods. This is necessary but not sufficient: passing does not guarantee that parallel trends hold post-treatment.

9.2.2.5 Estimation

With panel data,

$$\widehat{\text{ATT}} = \underbrace{\frac{1}{N_1}\sum_{i=1}^{N} G_i\{Y_{i1} - Y_{i0}\}}_{\text{average trend, treated}} - \underbrace{\frac{1}{N_0}\sum_{i=1}^{N}(1-G_i)\{Y_{i1} - Y_{i0}\}}_{\text{average trend, control}},$$

with standard errors from the usual difference-in-means estimator. Equivalent regression: with $\Delta Y_i = Y_{i1} - Y_{i0}$, $\Delta Y_i = \alpha + \tau G_i + \varepsilon_i$ gives $\hat{\tau}_{\text{OLS}} = \widehat{\text{ATT}}$ (OLS on a binary regressor equals the difference in group means), convenient for adding pre-treatment covariates. DiD can also be used with repeated cross-sections.

Card and Krueger with numbers (average full-time-equivalent employment per restaurant):

	Before (Feb 1992)	After (Nov 1992)	Difference
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD estimate			+2.76

DiD $= (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$: no evidence of employment loss from the minimum-wage increase.

9.2.2.6 DiD and two-way fixed effects (2×2)

The simplified TWFE model for two groups and two periods is

$$Y_{it} = \mu + \gamma G_i + \delta t + \tau X_{it} + \varepsilon_{it},$$

with fixed effects capturing all time-invariant group differences (γG_i) and unit-invariant time effects (δt). Since $X_{it} = G_i \cdot t$, the four cell means are μ, $\mu + \gamma$, $\mu + \delta$, $\mu + \gamma + \delta + \tau$, and the coefficient τ equals the DiD estimate.

9.2.3 Conditional and nonlinear DiD

Motivation. Weaken parallel trends with covariates, and replace mean trends by rank preservation.

9.2.3.1 Conditional parallel trends

Unconditional parallel trends may fail if groups differ in observed covariates X_i. *Conditional parallel trends*: for all x,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1, X_i = x] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0, X_i = x].$$

This is weaker: trends need only be parallel within covariate strata (states with different demographics may trend differently, but states with similar demographics should evolve similarly). Estimators:

- **Outcome regression:** include X_i in the (TWFE) model.
- **IPW-DiD:** reweight control units by $\hat{e}(X_i)/(1-\hat{e}(X_i))$ to match the treated group's covariate distribution.
- **Doubly robust DiD:** combine both; consistent if either the outcome model or the propensity model is correct (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021), with the same logic as DR estimation in Chapter 6.

9.2.3.2 Nonlinear DiD (changes-in-changes)

Standard DiD's parallel trends in levels are not preserved under log, $\sqrt{\cdot}$, etc. Key insight: assume parallel trends in *ranks* (quantiles) instead of levels: absent treatment, a unit at the q-th quantile of its group in period 0 remains at the q-th quantile in period 1. Its value changes, but its rank is preserved. This rank-preservation assumption is invariant to monotone transformations (the q-th quantile of Y maps to the q-th quantile of $\log Y$).

Formalization (Athey and Imbens 2006). Let $F_{gt}(y) = P(Y_{it}(0) \leq y \mid G_i = g)$. Observed: F_{00}, F_{01}, F_{10} (untreated outcomes), and the treated post-period CDF $F_{11}^{(1)}$ of $Y_{i1}(1)$. Goal: the counterfactual F_{11}. Quantile treatment effect:

$$\tau(q) = (F_{11}^{(1)})^{-1}(q) - F_{11}^{-1}(q).$$

Define the period-0-to-1 value shift at each rank: $h_0(y) = F_{01}^{-1}(F_{00}(y))$ for controls and $h_1(y) = F_{11}^{-1}(F_{10}(y))$ for the treated. *Identification assumption:* $h_0 = h_1$, i.e. the value shift at each rank is the same in both groups. Then $F_{11}^{-1}(F_{10}(y)) = F_{01}^{-1}(F_{00}(y))$, and solving,

$$F_{11}(y) = F_{10}(F_{00}^{-1}(F_{01}(y))).$$

(Take $y' = F_{01}^{-1}(F_{00}(y''))$; the assumption says $F_{11}(y') = F_{10}(y'')$ with $y'' = F_{00}^{-1}(F_{01}(y'))$.)

9.2.3.3 Key takeaways

- Parallel trends are the core of DiD identification.
- DiD handles time-invariant unobserved confounding but is vulnerable to time-varying confounders (Ashenfelter's dip).
- Conditional parallel trends with covariates lead to IPW-DiD and doubly robust DiD.
- Pre-trend tests are useful but do not validate parallel trends; sensitivity analysis is preferred.
- Linear DiD is not invariant to nonlinear transformations; nonlinear DiD needs distributional assumptions.
- TWFE gives the same result in the 2×2 case but needs care with staggered timing.

9.2.4 Appendix: Pre-testing bias

Common practice: run a pre-trend test and proceed with DiD only if it is insignificant. Problem: conditioning on passing biases the DiD estimate. Tests have low power against small pre-trends and often fail to reject; "passing" selects samples in which pre-trends happen to look flat. The same chance variation that hid the pre-trend persists into the post-period and distorts the estimate. Result: under-rejection of the pre-trend test leads to over-rejection of "no treatment effect". A passed pre-trend test gives comfort, not validation (Roth 2022).

9.2.5 Appendix: Sensitivity analysis for parallel trends

Parallel trends are untestable; how robust are the estimates? Allow bounded violations: let M bound the deviation

$$\left|\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]\right| \leq M.$$

For each M construct a confidence set for the ATT; as M grows the set widens. Report the *breakdown value*, the smallest M at which the set first includes 0 (the finding is no longer robust). One can also extrapolate from pre-treatment trend differences to choose plausible bounds (Rambachan and Roth 2023).

9.2.6 Appendix: Staggered treatment timing

Units often enter treatment at different times. Standard TWFE with staggered timing can be biased. The Goodman-Bacon decomposition shows TWFE is a weighted average of all possible 2×2 DiD comparisons: early-treated vs. never-treated (good), late-treated vs. never-treated (good), and late-treated vs. early-treated (dangerous: the early-treated serve as controls). If effects change over time, early-treated units are not valid controls for the late-treated, since their outcomes already reflect treatment; using them attenuates or even reverses the estimate.

Modern solutions. Callaway and Sant'Anna (2021): compute group-time average treatment effects using only never-treated or not-yet-treated units as controls. Event-study plots: plot effects by time relative to treatment onset, $t - t^*$, allowing visual inspection of pre-trends and dynamic effects. (See the lecture on staggered DiD.)

9.3 Staggered difference-in-differences

Motivation. The central question is how to estimate treatment effects with DiD when units adopt treatment at different times and effects may vary across units and over time, a setting in which the usual two-way fixed-effects regression can be badly misleading.

9.3.1 Staggered adoption

Motivation. Describe the setting and why it complicates DiD.

The canonical DiD has two periods and two groups. Much applied work has more periods, with units becoming treated at different times: e.g. group A treated early, group B later, group C later still, and a never-treated group. Plots of staggered DiD without and with treatment reversal show outcome paths of groups A, B, C over 15 periods: after adoption the treated paths jump and may keep changing (dynamic effects) relative to their counterfactual (triangles); with treatment reversal, effects switch on and off.

Staggered DiD generalizes DiD to settings where treatment timing varies, common when policies are rolled out gradually across regions: a *staggered adoption design* (SAD). Units not yet treated can serve as controls for those already treated, so staggered adoption blurs the distinctions treatment/control and pre/post. Once treated, a unit remains treated (no reversal).

9.3.1.1 TWFE as is is not sufficient

The most common approach is the TWFE regression $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$. It works when (i) effects are not heterogeneous, or (ii) there are only two periods: then, under parallel trends and no anticipation, τ equals the ATT. But effects are very likely heterogeneous, varying across units or over time. With staggered timing, already-treated units exhibit treatment dynamics and act as controls; these dynamics leak into τ, making it very hard to interpret causally. The fix: use only "good" comparisons between groups instead of all possible comparisons.

9.3.1.2 Terminology

- **Static effects:** constant across post-treatment periods (e.g. a persistent 5% increase).
- **Dynamic effects:** change with time since adoption (grow, decay, fluctuate).
- **Homogeneous effects:** identical across units.
- **Heterogeneous effects:** vary across units (group- or unit-level differences) but are constant over time for each unit.
- **Carryover effects:** a past treatment keeps influencing outcomes after it is withdrawn; a specific form of dynamic effect.

9.3.1.3 Valid and invalid comparisons

At time t, a treated unit can be compared with never-treated units, not-yet-treated (later-treated) units, or already-treated (earlier-treated) units.

- **Valid:** treated vs. never-treated; treated vs. not-yet-treated.
- **Invalid:** treated vs. already-treated; later-treated vs. earlier-treated using the earlier group's post-treatment periods.

Bias arises when effects vary over time (dynamic) or across units (heterogeneous). (The Goodman-Bacon decomposition in Section 9.2 shows TWFE averages all 2×2 comparisons, including the invalid ones.)

Modern alternatives to TWFE modify the set of effective comparison units: Callaway and Sant'Anna (2021), group-time ATT estimators; Sun and Abraham (2021), event-study designs with proper controls; stacked DiD, separate DiDs per cohort to avoid contamination. They differ in which observations serve as comparisons and how covariates enter.

9.3.2 Callaway–Sant'Anna group-time ATTs

Motivation. Define, identify and aggregate effects by adoption cohort and calendar time.

9.3.2.1 Notation

G_i: the period in which unit i becomes treated (groups are defined by adoption time). $Y_{it}(0)$: untreated potential outcome in period t. $Y_{it}(g)$: potential outcome in period t if first treated in

period g. Y_{it}: observed outcome. Never-treated units have $Y_{it} = Y_{it}(0)$ for all t; for others

$$Y_{it} = \mathbb{1}\{G_i > t\}Y_{it}(0) + \mathbb{1}\{G_i \leq t\}Y_{it}(G_i) :$$

untreated outcomes before participation and treated outcomes (possibly depending on adoption time) afterwards. This notation embeds a no-anticipation assumption, which can be relaxed. C_i: indicator of the never-treated (control) group. D_{it}: indicator that i has been treated by t. X_i: pre-treatment covariates.

Staggered adoption assumption: for $t = 1, \ldots, T-1$, $D_{it} = 1 \Rightarrow D_{i,t+1} = 1$: once treated, units remain treated (they do not "forget" their treatment). Effects may be heterogeneous across time, groups, treatment sequences, etc.

9.3.2.2 Parallel trends

Based on never-treated units: for all $g = 2, \ldots, T$ and $t = 2, \ldots, T$ with $t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1].$$

It uses the never-treated for all eventually-treated groups, presuming (i) a large enough never-treated group exists and (ii) these units are similar enough.

Based on not-yet-treated units: for all g and s, t with $s \geq t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D_s = 0, G \neq g].$$

Units not yet treated by time $s \geq t$ can serve as comparisons when computing effects for the group first treated at g. This generally uses more data, but restricts some pre-treatment trends across groups (Marcus and Sant'Anna 2021): no free lunch. (Why is the restriction placed on s relative to t? Because $s \geq t$ guarantees, by staggered adoption, that comparison units are untreated in both periods $t-1$ and t.)

9.3.2.3 Group-time ATT and identification

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g],$$

the average effect for the group first treated in g, at time t. With three periods, $\mathrm{ATT}(g = 2, t = 3)$ is the effect in period 3 for units first treated in period 2.

Under either parallel-trends assumption, group-time ATTs are identified. With never-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1];$$

with not-yet-treated controls, for $t \geq g$,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0, G \neq g].$$

Derivation (never-treated case): $\mathbb{E}[Y_t(g) \mid G = g] = \mathbb{E}[Y_t \mid G = g]$ by the observation rule. Telescoping parallel trends from g to t, $\mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid C = 1] = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$; and $Y_{g-1}(0) = Y_{g-1}$ for group g (no anticipation). Combine. Each $\mathrm{ATT}(g, t)$ is a clean 2×2 DiD between group g and a valid comparison group, with base period $g - 1$.

9.3.2.4 Conditional parallel trends

Parallel trends are often more plausible conditional on pre-treatment covariates. *Never-treated:* for $t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$. *Not-yet-treated:* for $s \geq t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, D_s = 0, G \neq g]$. These allow covariate-specific trends, important when covariate distributions differ across groups (estimated by outcome regression, IPW or doubly robust DiD, as in Section 9.2).

9.3.2.5 Aggregation

By group. The average effect of participation for group g:

$$\theta_S(g) = \frac{1}{T-g+1} \sum_{t=2}^{T} \mathbb{1}\{g \leq t\} \, \text{ATT}(g,t),$$

highlighting heterogeneity with respect to adoption time. Aggregating further,

$$\theta_S^O := \sum_{g=2}^{T} \theta_S(g) \, P(G=g)$$

is the overall effect of participating across all groups that ever participate: close to a multi-period analogue of the two-period ATT, and the recommended single summary parameter.

By exposure length (event study). To highlight dynamics,

$$\theta_D(e) := \sum_{g=2}^{T} \mathbb{1}\{g+e \leq T\} \, \text{ATT}(g,g+e) \, P(G=g \mid g+e \leq T),$$

the average effect for units exposed to treatment for exactly e periods.

9.3.2.6 Summary

- Staggered DiD uses group-time ATTs $\text{ATT}(g,t)$: the effect in period t for units first treated in period g.
- It allows flexible clean controls: not-yet-treated or never-treated units.
- It accommodates pre-treatment (static) covariates via conditional parallel trends.
- Anticipation can be handled by shifting the base period before any behavioral adjustment.

References: Athey and Imbens (2022); Baker, Larcker and Wang (2022); Callaway and Sant'Anna (2021); Chiu, Lan, Liu and Xu (2023); Goodman-Bacon (2021); Wing et al. (2024).

9.4 Synthetic control

Motivation. The central question is how to estimate the effect of an intervention on a *single* treated unit (a state or country) when no single untreated unit is a credible comparison, by building a weighted combination of untreated units that tracks the treated unit before the intervention.

9.4.1 From DiD to synthetic controls

Motivation. Motivate a data-driven comparison unit when there is only one treated unit.

DiD works well with two groups and parallel trends. But with a single treated unit (a country adopting a policy), which control should be used? Any single control may not be comparable. The **synthetic control method** (SCM) constructs an optimal weighted combination of control units to approximate the counterfactual.

Abadie and Gardeazabal (2003) used a DiD-like approach for "quantitative case studies": an intervention in one country/state at one point in time; one treated group and many controls; compare time series of the treated unit with a control. Since there are many possible control choices, use a *convex combination* of the controls, a synthetic control, with weights minimizing pre-treatment differences between the treated unit and its synthetic counterpart.

9.4.1.1 Case study: California Proposition 99

In 1988 California passed Proposition 99, a \$0.25 per pack cigarette tax increase (the largest state tax increase at that time). Goal: the effect on per-capita cigarette sales. No single state is a good comparison. SCM builds a "synthetic California" from 38 control states; key donors are Utah (33.4%), Nevada (23.4%), Montana (19.9%) and Colorado (16.4%). Weights are chosen to match pre-1988 cigarette sales, income, beer consumption, retail price and the share of population aged 15–24. Result: by 2000 California's per-capita sales were about 26 packs lower than synthetic California's (Abadie, Diamond and Hainmueller 2010). In the figure, actual and synthetic California track each other closely from 1970 to 1988 and diverge afterwards. Without synthetic controls, California compared with the average of the other states differs already before 1988; the synthetic control reproduces California's pre-1988 smoking and predictor values, which, like matching, forces the researcher to demonstrate the affinity between the treated unit and its counterpart and guards against "extreme counterfactuals" far outside the convex hull of the data.

9.4.2 The estimator

Motivation. Define the setting, estimand and weights.

9.4.2.1 Setup

Time period	1	2	$\cdots$	T_0	T_0+1	$\cdots$	T
Treated unit ($i=1$)	0	0	0	0	1	1	1
Control group ($i=2,\dots,J+1$)	0	0	0	0	0	0	0

All units are untreated for T_0 periods; unit 1 is treated from T_0+1 to T. Potential outcomes $Y_{it}(x)$, $x \in \{0,1\}$; observed $Y_{it} = Y_{it}(X_{it})$. Estimand: the effect on the treated unit over time,

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \qquad t > T_0.$$

9.4.2.2 Missing counterfactuals and synthetic weights

By consistency, for $t > T_0$, $\tau_{1t} = Y_{1t} - Y_{1t}(0)$, so the missing $Y_{1t}(0)$ must be imputed. Choose weights $(w_2,\dots,w_{J+1})$ with

$$w_j \geq 0, \qquad \sum_j w_j = 1 \quad \text{(convex)},$$

minimizing, over pre-treatment periods $t \leq T_0$,

$$\Big\| Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \Big\| \quad \text{and} \quad \Big\| Z_1 - \sum_{j=2}^{J+1} w_j Z_j \Big\|,$$

where Z_i are pre-treatment predictors. The objectives are combined via a weighted norm with predictor-importance weights V (Appendix 9.4.6); a penalty on the dispersion of the weights can be added. The hope is that, for $t > T_0$, $\sum_{j=2}^{J+1} w_j Y_{jt} \approx Y_{1t}(0)$, giving $\hat{\tau}_{1t} = Y_{1t} - \sum_j w_j Y_{jt}$.

9.4.2.3 Model-based justifications

Model 1: interactive factor model.

$$Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it},$$

with θ_t time-varying coefficients on covariates, α_i a unit fixed effect, δ_t a common time shock, λ_t a $1 \times F$ vector of common factors (unobserved macro shocks) and μ_i an $F \times 1$ vector of factor loadings (each unit's sensitivity to those shocks). The term $\lambda_t \mu_i$ allows time-varying confounding in a structured way; DiD is the special case without it.

Model 2: autoregressive model (no fixed effects):

$$Y_{i,t+1}(0) = \alpha_t Y_{it}(0) + \beta_{t+1}^{\top} W_{i,t+1} + u_{i,t+1}, \qquad W_{i,t+1} = \gamma_t Y_{it}(0) + \Pi_t W_{it} + v_{i,t+1},$$

with time-varying covariates W_{it}, AR coefficient α_t (the dynamic source replacing the factor structure), coefficients β_{t+1}, feedback γ_t from outcome to covariates, and autoregressive matrix Π_t.

Properties. Suppose perfect balancing weights exist: $\sum_j w_j Y_{jt} = Y_{1t}$ for $t \leq T_0$ and $\sum_j w_j Z_j = Z_1$. Let $\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}$ for $t > T_0$.

- Under Model 1, $\hat{Y}_{1t}(0) \to Y_{1t}(0)$ as $T_0 \to \infty$: consistent, improving as the pre-intervention period grows (matching many pre-period outcomes forces $\sum_j w_j \mu_j \approx \mu_1$).
- Under Model 2, $\mathbb{E}[\hat{Y}_{1t}(0)] = \mathbb{E}[Y_{1t}(0)]$ (given perfect fit): unbiased even with a single pre-treatment period.

9.4.2.4 Convexity: no extrapolation

SCM requires $w_j \geq 0$ and $\sum_j w_j = 1$, so the synthetic control is a convex combination of donors. Example: California's pre-treatment smoking is 120 packs per capita, while all donors lie between 60 and 110; convex weights can reach at most 110, so the imperfect fit is visible and transparent. OLS, in contrast, may assign negative weights (e.g. $w_j = -0.3$); the weights still sum to one, but the synthetic outcome can lie far outside the observed range, and this extrapolation is hidden. SCM trades flexibility for transparency: one can always check whether the treated unit lies inside the convex hull of the donors.

Sparsity: geometric intuition. SC weights are sparse: most $w_j = 0$. Geometrically, the treated unit is approximated by a point on the boundary of the convex hull of donors, so only a few "vertex" donors get positive weight, which aids interpretation. All weight decisions are made without post-treatment data, protecting against specification searching.

9.4.2.5 Practical concerns

- To reduce interpolation bias and overfitting, restrict the donor pool to units similar to the treated unit.
- Donors must not be affected by the treatment (no interference or spillovers).
- "Although we define our synthetic controls as convex combinations of unexposed units, negative weights or weights larger than one can be used at the cost of allowing extrapolation." (Relaxing the no-extrapolation property.)
- Convex weights limit flexibility: the treated unit's pre-treatment outcomes may not be matched perfectly, especially if it lies outside the convex hull; the trade-off is deliberate, for interpretability and robustness.
- With $N = 1$ treated unit, inference uses placebo tests.

9.4.3 Inference and bias

Motivation. Assess significance by placebo permutation, and understand bias from imperfect fit.

9.4.3.1 Placebo tests

With one treated unit, sampling-based inference is unavailable. Iteratively apply SCM to each control state as if it were treated: for each of the 38 control states, build a synthetic control and estimate a "placebo effect". If SCM works well, most placebo effects are near zero. This is a permutation/randomization-inference argument: under the null of no effect, California's gap should look like a typical draw from the distribution of placebo gaps; an unusually large gap is evidence of a real effect. The "p-value" is the fraction of placebo gaps at least as large as California's. In Abadie et al. (2010) only 1 of 34 states had a larger gap, $p \approx 0.03$.

In the placebo plot, gray lines show gaps $Y_{jt} - \hat{Y}_{jt}^{\text{SC}}$ for control states and a red line California's gap: most placebo gaps are small and centered at zero, while California's diverges sharply after 1988. Refinement: exclude states with poor pre-treatment fit (high pre-treatment RMSPE), whose placebo gaps are noisy and uninformative.

9.4.3.2 Why imperfect pre-treatment fit causes bias

Under Model 1, if $\sum_j w_j Y_{jt} = Y_{1t}$ for all $t \le T_0$, the factor loadings are approximately matched, $\sum_j w_j \mu_j \approx \mu_1$. If the match is imperfect for some $t \le T_0$, then $\sum_j w_j \mu_j \neq \mu_1$, and post-treatment the common factors create a systematic wedge:

$$\sum_j w_j Y_{jt}(0) - Y_{1t}(0) \approx \lambda_t \Big(\sum_j w_j \mu_j - \mu_1 \Big) \neq 0,$$

the other terms ($Z_i^\top \theta_t$, α_i, δ_t) cancelling by predictor balancing and ε_{it} averaging out. Intuition: the synthetic control does not capture unit 1's response pattern to macro shocks, so predictions drift after treatment. This is especially problematic when the treated unit lies outside the convex hull of donors.

9.4.3.3 Bias correction: augmented SCM

Use an outcome model to correct residual mismatch (Ben-Michael, Feller and Rothstein 2021). Model $Y_{it}(0) = m_{it} + \varepsilon_{it}$; fit the regression on control units only ($j = 2, \dots, J+1$) and let $\hat{m}_{it}$ denote predictions, including for $i = 1$. The augmented estimator is

$$\hat{Y}_{1t}^{\text{aug}}(0) = \underbrace{\sum_{j=2}^{J+1} w_j Y_{jt}}_{\text{SC estimate}} + \underbrace{\Big(\hat{m}_{1t} - \sum_{j=2}^{J+1} w_j \hat{m}_{jt} \Big)}_{\text{bias correction}} = \underbrace{\hat{m}_{1t}}_{\text{outcome model}} + \underbrace{\sum_{j=2}^{J+1} w_j (Y_{jt} - \hat{m}_{jt})}_{\text{SC-weighted residuals}}.$$

Covariates are easy to add. It is doubly robust: consistent if either the SC weights or the outcome model is approximately correct (analogous to AIPW and doubly robust DiD).

9.4.4 Discussion

Motivation. Weigh strengths and limitations and position SCM among alternatives.

Strengths	Limitations
Transparency: weights are explicit; one can inspect which donors contribute and how much.	Needs a good donor pool: similar units with no spillovers.
Data-driven: no need to pick a single comparison unit by hand.	Designed for a single treated unit; extensions needed for several.
No parallel-trends assumption (uses a factor model instead).	Sensitive to donor-pool and predictor choice.
Pre-treatment fit is a built-in diagnostic of credibility.	Formal inference is limited (permutation-based, not sampling-based); convexity can prevent good fit when the treated unit is an outlier.

(Abadie 2021.)

When to use SCM: one or very few treated units with multiple controls; a rich pre-treatment period; parallel trends implausible but a factor structure reasonable. Consider DiD with multiple treated and control units and plausible parallel trends; consider synthetic DiD to combine both, especially in larger panels. Open research areas: formal inference beyond permutation tests (conformal inference), sensitivity analysis for SCM specifications, experimental design with synthetic controls, external validity.

Extensions. Augmented SCM (bias correction, double robustness); synthetic difference-in-differences (Arkhangelsky et al. 2021: unit weights as in SCM, time weights and unit fixed effects as in DiD, valid large-panel inference); penalized SCM (Abadie and L'Hour 2021: relax strict convexity with L_1/L_2 penalties, allowing regularized partial extrapolation); conformal inference for SCM (Chernozhukov, Wüthrich and Zhu 2021: distribution-free prediction intervals via permutation of residuals); matrix completion (Athey et al. 2021: the counterfactual as a missing-data problem with a low-rank approximation).

DiD versus synthetic control.

	Difference-in-differences	Synthetic control
Setup	two groups, two (or more) periods	one treated unit, many donors
Key assumption	parallel trends	pre-treatment fit (factor model)
Estimand	ATT (group level)	τ_{1t} (unit level)
Strength	simple, transparent	handles one treated unit well
Weakness	sensitive to nonlinear transforms	requires a good donor pool
Extensions	staggered DiD, nonlinear DiD	augmented SCM, matrix completion

9.4.5 Appendix: More treated units

Two estimation approaches generalize to any number of treated units. *Interactive fixed effects:* with $Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it}$ (Model 1), estimate the IFE directly instead of weighting, by iterating:

(1) treat the IFE terms as fixed and fit the parametric part on untreated observations to get $\hat{\theta}$;

(2) treat the covariate coefficients as fixed and estimate the IFE terms by factor analysis (principal components of residuals);

(3) repeat until convergence. *Matrix completion* (Athey et al. 2021): treat the matrix of control potential outcomes $\mathbf{Y}(0)$ as partially missing and estimate a low-rank matrix $\mathbf{L}$ approximating its observed entries, with regularization (as in recommendation systems).

9.4.6 Appendix: Predictor importance weights V

The unit weights minimize a V-weighted predictor mismatch:

$$w(V) = \underset{w\geq 0,\ \sum_j w_j = 1}{\arg\min} \sum_{m=1}^{k} v_m \Big(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \Big)^2,$$

where $v_m \geq 0$ says how much predictor m (a covariate or a pre-period outcome) matters. Simple choices: equal weights $v_m = 1$; variance scaling $v_m = 1/\operatorname{Var}(X_m)$ (common scale); regression-based $v_m \propto |\hat{\beta}_m|$ from an outcome regression on donors. Abadie's standard, automatic choice is nested optimization: $V^* = \arg\min_V \mathrm{MSPE}_{\mathrm{pre}}(w(V))$, the outer loop choosing V to minimize pre-treatment outcome MSPE and the inner loop solving for $w(V)$.

Caveats. (1) *Outcome-lag collapse* (Kaul, Klößner, Pfeifer and Schieler 2022): including all pre-period outcome lags drives $v_m \to 0$ for the other predictors, so matching collapses to outcomes only and covariates are ignored; use only a few lags (e.g. every five years).

(2) *Sensitivity:* V depends on the predictor list and lag choice, and different specifications give different donor weights. Practice: choose predictors theoretically related to the outcome; check robustness across several V methods (equal, variance, nested) and report whether the effect is stable; report donor weights explicitly so readers see which donors drive the result.

9.5 Synthetic difference-in-differences

Motivation. The central question is how to combine the strengths of difference-in-differences (invariance to additive unit-level shifts) and synthetic control (re-weighting units to match pre-treatment trends) into one estimator for panel data.

9.5.1 DiD, SC and SDiD

Motivation. Recall DiD and SCM and preview what SDiD changes.

Picture (California Proposition 99, cigarette consumption 1970–2000). *DiD* compares California with the unweighted average of control states, using all pre-periods equally; the level difference is allowed, but the trends must be parallel, and visibly they are not. *SC* re-weights control states so that the synthetic control matches California's level and trend before 1988, using no time weights. *SDiD* re-weights control states so that their trend (not necessarily level) is parallel to California's, and re-weights pre-treatment years, concentrating weight on the few years just before treatment (the shaded time weights).

Recap. DiD estimates effects by comparing outcome changes over time between treated and control groups, under parallel trends (a common trend absent treatment) while allowing time-invariant level differences. SCM addresses comparative case studies with a single treated unit, building a matched synthetic control from many donors. Both are for panel data and similar in spirit, but differ:

(1) SCM adds unit weights;

(2) SCM has time fixed effects but no unit fixed effects and no overall intercept.

SDiD (Arkhangelsky et al. 2021). Like SCM, it re-weights and matches pre-treatment trends, reducing reliance on parallel trends. Like DiD, it is invariant to additive unit-level shifts.

9.5.2 Formulations

Motivation. Write all three estimators as (weighted) TWFE regressions.

With unit fixed effects α_i, time fixed effects β_t, an intercept μ and treatment indicator X_{it}, the effect τ is estimated by

$$\hat{\tau}^{\text{DiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2, \tag{9.1}$$

$$\hat{\tau}^{\text{SCM}} = \arg\min_{\tau,\mu,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i, \tag{9.2}$$

$$\hat{\tau}^{\text{SDiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i\hat{\lambda}_t. \tag{9.3}$$

The unit fixed effects give invariance to additive unit-level shifts; the weights $\hat{\omega}_i$ (and $\hat{\lambda}_t$) match pre-treatment trends.

9.5.2.1 Setting and algorithm

A balanced panel with N units and T periods, outcomes Y_{it}, binary exposure $X_{it} \in \{0,1\}$. The first N_{co} units are never exposed; the last $N_{tr} = N - N_{co}$ units are exposed after T_{pre} ($T_{post} = T - T_{pre}$). Like SCM, SDiD finds unit weights $\hat{\omega}$ aligning pre-exposure trends of unexposed units with those of exposed units; it also finds time weights $\hat{\lambda}$ balancing pre-exposure periods with post-exposure ones.

1. Compute unit weights $\hat{\omega}$.
2. Compute time weights $\hat{\lambda}$.
3. Compute $(\hat{\tau}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i,t}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\hat{\omega}_i\hat{\lambda}_t$.

9.5.2.2 Unit weights

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0\in\mathbb{R},\ \omega\in\Omega}{\arg\min} \left\|\bar{y}_{pre,tr} - (\omega_0 + \omega_{co}^{\top}Y_{pre,co})\right\|_2^2 + \zeta^2 T_{pre}\|\omega_{co}\|_2^2,$$

where $\bar{y}_{pre,tr} = \frac{1}{N_{tr}}\sum_{i=N_{co}+1}^{N} Y_{it}$ is the average pre-period outcome path of the treated, $\omega_{co}^{\top}Y_{pre,co} = \sum_{i=1}^{N_{co}} \omega_i Y_{it}$ is the synthetic control path, and Ω is the simplex (non-negative weights summing to one). Differences from SCM:

1. SDiD allows an intercept ω_0: the synthetic control need not have the same level as the treated units, only a *parallel trend*.
2. The L_2 penalty spreads non-zero weights across control units, instead of the few contributors typical of SCM, forcing more units to be used.

Regularization (optional). $\zeta = (N_{tr}T_{post})^{1/4}\,\hat{\sigma}(\Delta_{it})$, where $\Delta_{it} = Y_{i(t+1)} - Y_{it}$ is the first difference (temporal fluctuation) of control outcomes in the pre-period,

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}\Delta_{it}, \qquad \hat{\sigma}^2(\Delta_{it}) = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}(\Delta_{it} - \bar{\Delta})^2,$$

the average fluctuation and the variance of fluctuations.

9.5.2.3 Time weights

$\hat{\lambda}$ minimizes the difference between pre- and post-treatment periods for the controls:

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0\in\mathbb{R},\ \lambda\in\Lambda}{\arg\min} \left\|\bar{y}_{post,co} - (\lambda_0 + \lambda_{pre}^{\top}Y_{pre,co})\right\|_2^2,$$

where $\bar{y}_{post,co} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}$ is a $1 \times N_{co}$ row vector of post-period averages per control unit, $Y_{pre,co}$ is the $T_{pre} \times N_{co}$ matrix of pre-period control outcomes, λ_{pre} a $1 \times T_{pre}$ vector (one entry per pre-period, in the simplex Λ), and λ_0 an intercept allowing the post-period to lie above or below all pre-periods, as in many applications with a clear trend.

9.5.3 Comparisons and takeaways

Motivation. Explain what the weights and fixed effects buy.

SDiD versus DiD. DiD solves the same TWFE problem without unit or time weights. The weights make the regression "local": it emphasizes units similar on average to the treated in their past, and periods similar on average to the treated periods. Using only similar units and periods makes the estimator more robust, and the weights can improve precision by implicitly removing systematic, predictable parts of the outcome.

SDiD versus SCM. SCM omits the unit fixed effects and the time weights: $\hat{\tau}^{\text{SCM}} = \arg\min_{\mu,\beta,\tau} \sum_{i,t} (Y_{it} - \mu - \beta_t - X_{it}\tau)^2 \hat{\omega}_i^{\text{SCM}}$. Time weights can remove bias and improve precision by eliminating periods very different from the post-treatment periods. Unit fixed effects make the model more flexible (strengthening robustness) and often explain much of the outcome variation.

Key takeaways.

- Like SCM, SDiD works with multiple periods when pre-treatment trends are not parallel.
- Unlike SCM, its unit weights ω build a control that is only *parallel* to the treated group, not matched in level.
- Like DiD, it uses time (β) and unit (α) fixed effects, which explain outcome variance and reduce the estimator's variance.
- Its unit weights use an L_2 penalty (more spread out across controls) and allow an intercept ω_0.
- It introduces time weights, which neither DiD nor SCM has.